PH.48111 **FALL 2026**

Autonomous Theory Construction Laboratory

Lab 3

Garam Yun Due 2026-09-20 23:59

Topic

Grammar of Candidate Theories

Submission

Submit one ZIP archive containing report.pdf, the configuration file, and the minimal code needed to reproduce the run. The maximum file size is 25 MB.

Problems

1. Specify the input, output, and stopping condition for a minimal implementation of Grammar of Candidate Theories.
$$G ::= \Phi | \partial G | G \cdot G | f(G;\theta)$$
2. Implement one accepted and one rejected update. Preserve the data version, configuration, seed, and decision record.
$$dim(term) = dim(target)$$
3. Run a negative control, report the observed failure mode, and explain whether the acceptance rule behaved as intended.
$$gate(x) = 1/(1 + e^{-(x-x_0)/w})$$

Show enough intermediate work that another student can reproduce the calculation.